

One interface, four front ends

Each tinted column is one drive: two different physical devices on top, and the dashed ODOMETRY card below holding what both of them were reduced to. Badge “paper” marks the front end used in the paper’s experiments.

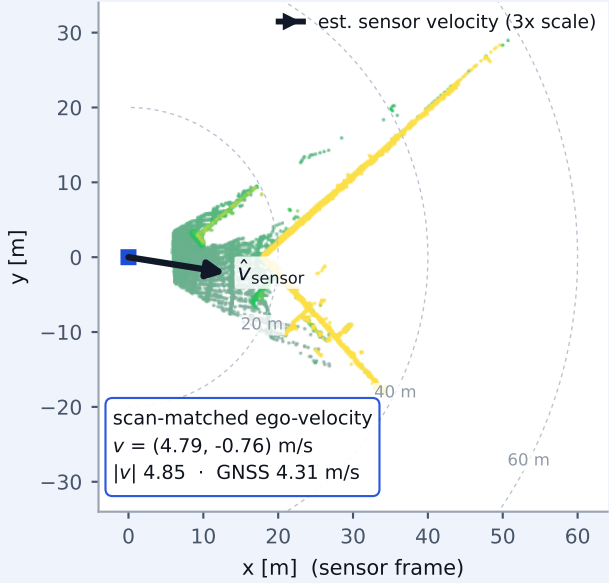
A2D2 · drive 20180810_150607

two devices · no shared field of view, clock or quantity

SENSOR A · LiDAR

paper

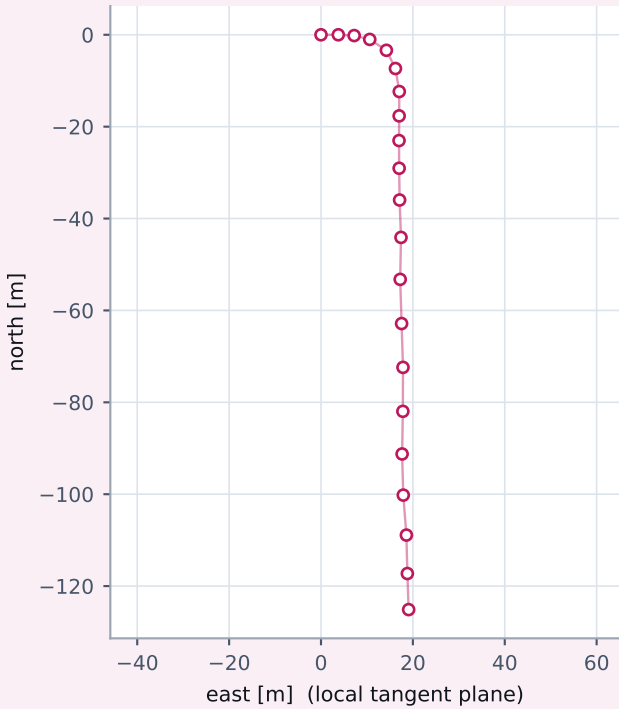
Velodyne VLP-16, FRONT_CENTER view · ≈ 30 Hz
range + bearing to surfaces $\rightarrow$ scan matching
shown: one scan, 14 000 points



SENSOR B · GNSS

demo

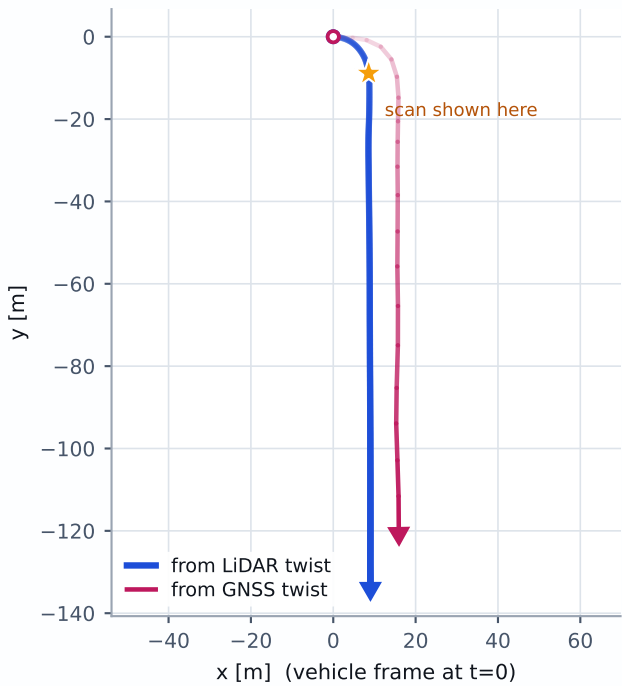
vehicle-bus GNSS, position only · 1 Hz fix
absolute position on Earth $\rightarrow$ differencing fixes
shown: 21 fixes at 1.0 Hz, no attitude



ODOMETRY

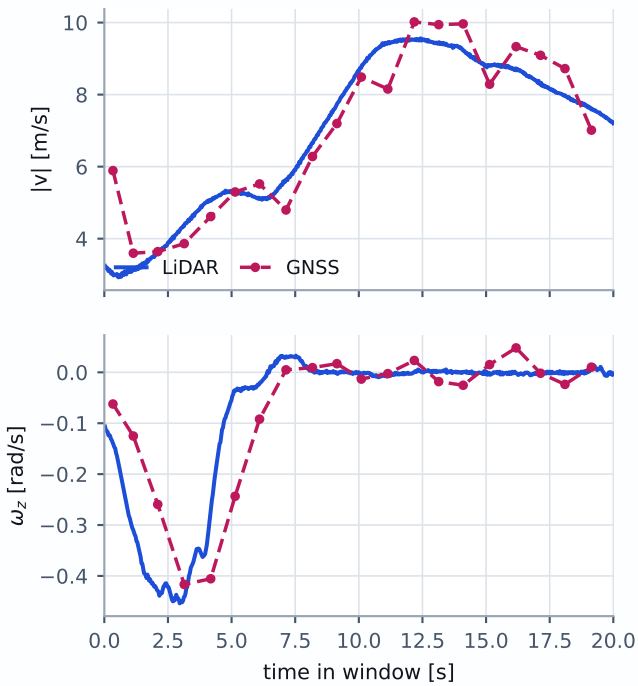
Trajectory from (v, ω)

139 m driven, -88° of heading



The twist, as delivered

path speed 6.95 vs 6.98 m/s · median ω gap 0.0242 rad/s



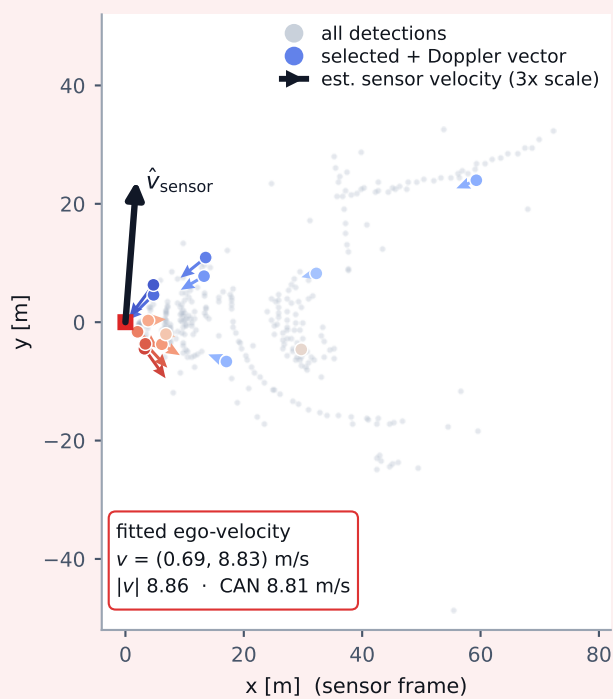
RadarScenes · sequence 1

two devices · no shared field of view, clock or quantity

SENSOR A · radar

paper

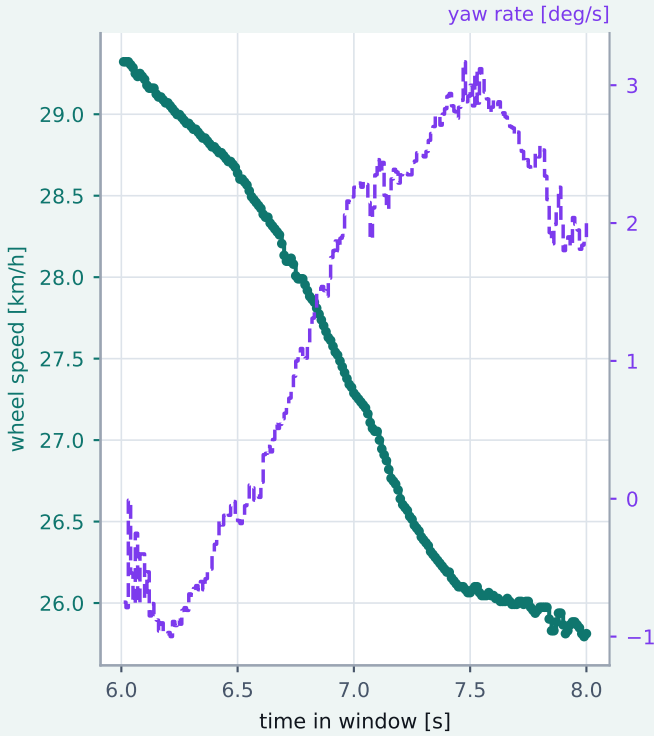
77 GHz series radar, sensor 1 · 13.5 scans/s
range, azimuth, Doppler $\rightarrow$ one-scan least squares
shown: one scan, 310 returns · 14 vectors



SENSOR B · CAN bus

demo

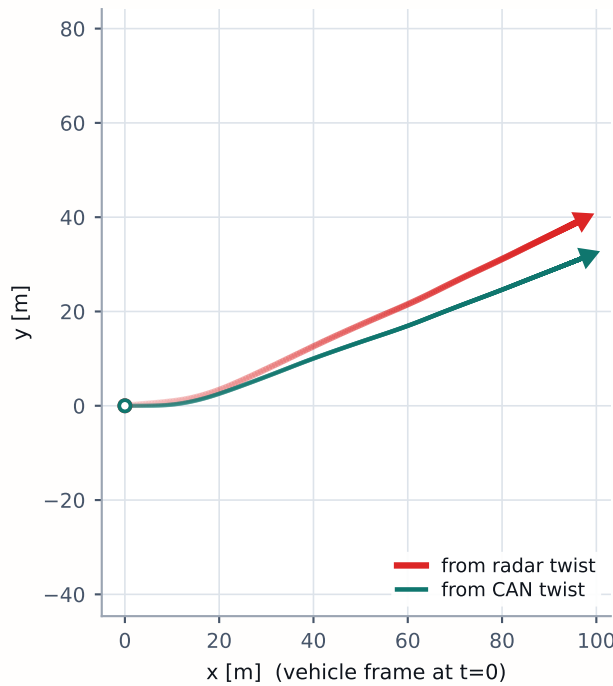
wheel encoders + yaw-rate gyro · 100 Hz
wheel rotation + yaw rate $\rightarrow$ vehicle kinematics
shown: 2 s zoom of the window



ODOMETRY

Trajectory from (v, ω)

105 m driven, 22° of heading



The twist, as delivered

median $|v|$ gap 0.078 m/s · one 2-D radar gives no ω

